

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Case Study

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 35%

Total Marks: 20

Code of Conduct

I ___ V.Tejaswini_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Case Study Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Total Marks Awarded:

Course Faculty

signature of student

Questions:

1. Retail Sales Data Analysis

A retail company has collected sales data containing customer details, product categories, quantities, prices, purchase dates, and cities from different branches. Using Python and Pandas, import the dataset, explore its structure and table functions, identify missing or unusual values, create suitable groupings based on product category and city, and analyze total sales and quantity sold. Present the important findings using appropriate charts and provide a brief interpretation of the results.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("retail_sales.csv")

print(df.head())

print(df.info())

print(df.describe())

print(df.isnull().sum())

df["Sales"] = df["Quantity"] * df["Price"]

print(df.groupby("Product_Category")["Sales"].sum())

print(df.groupby("City")["Sales"].sum())

print("Total Sales:", df["Sales"].sum())

print("Total Quantity:", df["Quantity"].sum())

df.groupby("Product_Category")["Sales"].sum().plot(kind="bar")

plt.title("Category-wise Sales")

plt.show()
```

```
First 5 rows:
  longitude  latitude  housing_median_age  total_rooms  total_bedrooms  population  households  median_income  median_house_value
0   -114.31    34.19             15.0      5612.0      1283.0      1015.0       472.0         1.4936         66900.0
1   -114.47    34.40             19.0      7650.0      1901.0      1129.0       463.0         1.8200         80100.0
2   -114.56    33.69             17.0       720.0       174.0       333.0       117.0         1.6509         85700.0
3   -114.57    33.64             14.0     1501.0       337.0       515.0       226.0         3.1917         73400.0
4   -114.57    33.57             20.0     1454.0       326.0       624.0       262.0         1.9250         65500.0

Dataset Shape:
(17000, 9)

Column Names:
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
      'total_bedrooms', 'population', 'households', 'median_income',
      'median_house_value'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17000 entries, 0 to 16999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   longitude              17000 non-null  float64
1   latitude               17000 non-null  float64
2   housing_median_age     17000 non-null  float64
3   total_rooms            17000 non-null  float64
4   total_bedrooms         17000 non-null  float64
5   population             17000 non-null  float64
6   households             17000 non-null  float64
7   median_income          17000 non-null  float64
8   median_house_value     17000 non-null  float64
dtypes: float64(9)
memory usage: 1.2 MB
```

2. Customer Behavior Analysis

An online shopping company wants to understand customer purchasing behavior using a dataset containing customer ID, age, gender, location, product category, order value, and purchase date. Import and explore the data, separate relevant customer groups, analyze purchasing patterns, identify relationships between customer characteristics and order value, and identify any significant outliers. Present the findings using suitable visualizations and explain how the analysis can help the company improve its marketing strategy.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("customer_behavior.csv")

print(df.head())

print(df.info())

print(df.describe())

print(df.isnull().sum())

print(df.groupby("Gender")["Order_Value"].mean())

print(df.groupby("Product_Category")["Order_Value"].sum())
```

```

print("Average Order Value:", df["Order_Value"].mean())

print("Correlation:", df["Age"].corr(df["Order_Value"]))

df["Order_Value"].plot(kind="box")

plt.title("Order Value Outliers")

plt.show()

df.groupby("Gender")["Order_Value"].mean().plot(kind="bar")

plt.title("Order Value by Gender")

plt.show()

```

```

0  longitude      17000 non-null  float64
1  latitude       17000 non-null  float64
2  housing_median_age  17000 non-null  float64
3  total_rooms     17000 non-null  float64
4  total_bedrooms  17000 non-null  float64
5  population      17000 non-null  float64
6  households      17000 non-null  float64
7  median_income   17000 non-null  float64
8  median_house_value 17000 non-null  float64
dtypes: float64(9)
memory usage: 1.2 MB
None

Statistical Summary:

```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households
count	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000
mean	-119.562108	35.625225	28.589353	2643.664412	539.410824	1429.573941	501.000000
std	2.005166	2.137340	12.586937	2179.947071	421.499452	1147.852959	384.000000
min	-124.350000	32.540000	1.000000	2.000000	1.000000	3.000000	1.000000
25%	-121.790000	33.930000	18.000000	1462.000000	297.000000	790.000000	282.000000
50%	-118.490000	34.250000	29.000000	2127.000000	434.000000	1167.000000	409.000000
75%	-118.000000	37.720000	37.000000	3151.250000	648.250000	1721.000000	605.000000
max	-114.310000	41.950000	52.000000	37937.000000	6445.000000	35682.000000	6082.000000

```

Missing Values:
longitude      0
latitude      0

```

3. Student Academic Performance

A college has collected student information including student ID, department, attendance percentage, assignment marks, internal marks, examination marks, and semester results. Explore the dataset using Pandas, create groups based on departments, identify correlations between attendance and academic performance, detect unusual performance values, and analyze the factors associated with higher marks. Present the results using charts and provide suitable recommendations to improve student performance.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("student_performance.csv")

print(df.head())

print(df.info())

print(df.describe())

print(df.isnull().sum())

df["Total_Marks"] = (
    df["Assignment_Marks"] +
    df["Internal_Marks"] +
    df["Examination_Marks"]
)

print(df.groupby("Department")["Total_Marks"].mean())

print("Correlation:")

print(df["Attendance_Percentage"].corr(df["Total_Marks"]))

df["Total_Marks"].plot(kind="box")

plt.title("Marks Outliers")

plt.show()

df.groupby("Department")["Total_Marks"].mean().plot(kind="bar")

plt.title("Department-wise Marks")

plt.show()
```

```

*** Statistical Summary:

```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households
count	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000
mean	-119.562108	35.625225	28.589353	2643.664412	539.410824	1429.573941	501.000000
std	2.005166	2.137340	12.586937	2179.947071	421.499452	1147.852959	384.000000
min	-124.350000	32.540000	1.000000	2.000000	1.000000	3.000000	1.000000
25%	-121.790000	33.930000	18.000000	1462.000000	297.000000	790.000000	282.000000
50%	-118.490000	34.250000	29.000000	2127.000000	434.000000	1167.000000	409.000000
75%	-118.000000	37.720000	37.000000	3151.250000	648.250000	1721.000000	605.000000
max	-114.310000	41.950000	52.000000	37937.000000	6445.000000	35682.000000	6082.000000

```

Missing Values:
longitude      0
latitude      0
housing_median_age  0
total_rooms    0
total_bedrooms  0
population     0
households     0
median_income  0
median_house_value  0
dtype: int64

Duplicate Rows:
0

```

4. Healthcare Patient Analysis

A hospital maintains patient records containing age, gender, diagnosis, treatment type, admission date, length of stay, and medical expenses. Import and explore the dataset, identify outliers in patient age and treatment expenses, group patients according to diagnosis and treatment type, and analyze relationships between length of stay and medical expenses. Create appropriate visualizations, including time-related charts where applicable, and prepare a concise presentation of the major findings.

Code:

```

import pandas as pd

import matplotlib.pyplot as plt

df = pd.read_csv("healthcare_patients.csv")

print(df.head())

print(df.info())

print(df.describe())

print(df.isnull().sum())

print(df.groupby("Diagnosis")["Medical_Expenses"].mean())

print(df.groupby("Treatment_Type")["Medical_Expenses"].mean())

```

```

print("Total Expenses:", df["Medical_Expenses"].sum())

print("Correlation:",

      df["Length_of_Stay"].corr(df["Medical_Expenses"]))

df["Medical_Expenses"].plot(kind="box")

plt.title("Medical Expense Outliers")

plt.show()

df.plot(

    x="Length_of_Stay",

    y="Medical_Expenses",

    kind="scatter"

)

plt.title("Stay vs Medical Expenses")

plt.show()

```

```

*** dtypes: float64(9)
memory usage: 1.2 MB
None

Statistical Summary:

```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households
count	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000
mean	-119.562108	35.625225	28.589353	2643.664412	539.410824	1429.573941	501.000000
std	2.005166	2.137340	12.586937	2179.947071	421.499452	1147.852959	384.000000
min	-124.350000	32.540000	1.000000	2.000000	1.000000	3.000000	1.000000
25%	-121.790000	33.930000	18.000000	1462.000000	297.000000	790.000000	282.000000
50%	-118.490000	34.250000	29.000000	2127.000000	434.000000	1167.000000	409.000000
75%	-118.000000	37.720000	37.000000	3151.250000	648.250000	1721.000000	605.000000
max	-114.310000	41.950000	52.000000	37937.000000	6445.000000	35682.000000	6082.000000

```

Missing Values:
longitude      0
latitude       0
housing_median_age  0
total_rooms    0
total_bedrooms 0
population     0
households     0
median_income  0
median_house_value 0
dtype: int64

Duplicate Rows:
0

```